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Association between HbA_{1c} and peripheral neuropathy in a 10-year follow-up study of people with normal glucose tolerance, impaired glucose tolerance and Type 2 diabetes

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What's new?

- This is the first 10-year follow-up of the association between HbA_{1c} and peripheral neuropathy in people with normal glucose tolerance, prediabetes and Type 2 diabetes.
- The results suggest an association between sural nerve amplitude and HbA_{1c} at all levels of HbA_{1c} and not only in those with diabetes.

Abstract

Aims To explore the association between HbA_{1c} and sural nerve function in a group of people with normal glucose tolerance, impaired glucose tolerance or Type 2 diabetes.

Methods We conducted a 10-year follow-up study in 87 out of an original 119 participants. At study commencement (2004), 64 men and 55 women (mean age 61.1 years) with normal glucose tolerance (*n*=39), impaired glucose tolerance (*n*=29), or Type 2 diabetes (*n*=51) were enrolled. At the 2014 follow-up (men, *n*=46, women, *n*=41; mean age 71.1 years), 36, nine and 42 participants in the normal glucose tolerance, impaired glucose tolerance and Type 2 diabetes categories, respectively, were re-tested. Biometric data and blood samples were collected, with an electrophysiological examination performed on both occasions.

Results At follow-up, we measured the amplitude of the sural nerve in 74 of the 87 participants. The mean amplitude had decreased from 10.9 µV (2004) to 7.0 µV (2014;

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P<0.001). A 1% increase in HbA_{1c} was associated with a ~1% average decrease in the amplitude of the sural nerve, irrespective of group classification. Crude and adjusted estimates ranged from –0.84 (95% CI –1.32, –0.37) to –1.25 (95% CI –2.31, –0.18). Although the mean conduction velocity of those measured at both occasions (*n*=73) decreased from 47.6 m/s to 45.8 m/s (*P*=0.009), any association with HbA_{1c} level was weak. Results were robust with regard to potential confounders and missing data.

Conclusions Our data suggest an association between sural nerve amplitude and HbA_{1c} at all levels of HbA_{1c}. Decreased amplitude was more pronounced than was diminished conduction velocity, supporting the notion that axonal degeneration is an earlier and more prominent effect of hyperglycaemia than demyelination.

Introduction

Peripheral neuropathy is a well-known long-term complication of Type 2 diabetes, occurring in up to 50% of people with Type 2 diabetes [1,2]. Symptoms of neuropathy are often present, even at the time of diagnosis, and affect ~8% of people with Type 2 diabetes [1,3–6]. Over a third of those with Type 2 diabetes with diabetic sensory-motor polyneuropathy develop neuropathic pain, which significantly impairs their quality of life [1]. Other debilitating symptoms of sensory-motor polyneuropathy include a loss of sensation and decreased sense of balance [7], which can lead to an increased risk of falls and tissue injury [7]. Combined with reduced blood flow, neuropathy in the feet increases the risk of foot ulcers, infection and the eventual need for limb amputation [8].

In contrast to Type 1 diabetes, in which hyperglycaemia is a prominent factor [9], the mechanisms that underlie the development of neuropathy in Type 2 diabetes remain obscure and multifactorial [9]. In Type 2 diabetes chronic inflammation, hyperglycaemia, obesity,

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hypertension, dyslipidaemia and insulin resistance are known factors relating to the onset and progression of diabetic neuropathy [2,10]. Nevertheless, levels of HbA_{1c} can predict sural nerve degeneration and regeneration in people with both Type 1 and Type 2 diabetes [11].

Currently, there is a lack of prospective cohort studies investigating the relationships between hyperglycaemia and nerve function among people with Type 2 diabetes. To our knowledge, no study has examined this relationship over time inclusive of people with impaired glucose tolerance (IGT) and normal glucose tolerance (NGT). In the study by Partanen *et al*. [12], the prevalence of aberrant nerve conduction (indicating demyelination) over a 10-year study period increased from 8% to 42% among people with Type 2 diabetes; abnormalities in control participants rose from 2% to 6% over the same period. Furthermore, the decreased nerve response amplitudes reported (indicative of axonal degeneration rather than demyelination) were even more pronounced [12].

Since that Finnish study, which was published over 20 years ago, there has been a trend towards greater obesity [13] and a higher prevalence of Type 2 diabetes [8], both increasing the risk of peripheral neuropathy. Moreover, treatment options and recommendations for Type 2 diabetes have been updated; therefore, contemporary data on this topic are now warranted. Our aim was to investigate the relationships between HbA_{1c} and electrophysiological nerve function in people with NGT, IGT or Type 2 diabetes.

Participants and methods

Study population

This study is a 10-year follow-up of a cohort originally from the population-based Västerbotten Intervention Programme [14], with participants recruited between November 2004 and April 2007; the original study population has been described elsewhere [15]. At the 2014 follow-up, six of the

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original 119 participants had died, and 26 declined further participation. The remaining 87 participants were included in the follow-up. The glycaemic status of individuals with NGT and IGT was verified by two standardized oral glucose tolerance tests [16,17] at baseline and at follow-up. NGT was defined as a capillary fasting plasma glucose concentration <6.1 mmol/l and a 2-h plasma glucose concentration <7.8 mmol/l; IGT was defined as fasting plasma glucose concentration < 7.0 mmol/l and a 2-h plasma glucose concentration ≥7.8 and <11.1 mmol/l. Lastly, Type 2 diabetes was defined as a fasting plasma glucose concentration of ≥7.0 mmol/l and/or a 2-h plasma glucose concentration ≥11.1 mmol/l, as defined by the 1999 WHO criteria [16,17]. Two participants in the 2004 NGT group, and six participants in the 2004 IGT group, had developed Type 2 diabetes by 2014. Further, four participants in the NGT group had developed IGT, and eight participants in the IGT group had normalized their glucose tolerance by 2014, as shown in Fig. 1.

All participants provided informed consent, and the regional ethical review board of Umeå University, Umeå, Sweden approved the study (ethical application no. Dnr 2013-21-31M).

Measurements

Participants were assessed using the same questionnaires, examinations, measurements and tests as were performed 10 years earlier [15]. Briefly, the assessment included a questionnaire on social status and previous medical history, including perceived hypo- or hypersensitivity to touch. Smoking or exposure to nicotine or other toxic substances were assessed. Height and weight were measured along with blood pressure using a standard sphygmomanometer in the supine position after a 10-min rest. Blood samples were drawn for analysis of HbA_{1c}. Mono S values from previous examinations were converted to International Federation of Clinical Chemistry (IFCC)-standardized values according to the formula HbA_{1c} (IFCC, mmol/mol)= HbA_{1c} (Mono S, %) × 10.45 – 10.62. HbA_{1c} values are reported in IFCC,

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mmol/mol (DCCT, %) throughout. Blood counts (haemoglobin, EPK, EVF, MCV, LPK, TRC) along with measurements of C-reactive protein, creatinine, total cholesterol and triglycerides were performed. GFR was calculated according to the Cockroft–Gault formula, whereby creatinine clearance = [(140–age]≤ mass (kg) × constant)/serum creatinine (in µmol/l)]. The constant is 1.23 for men and 1.04 for women.

To rule out vitamin B12 deficiency and hypothyroidism as causes of neuropathy, homocysteine and thyroid-stimulating hormone were measured.

Electrophysiological assessment

An experienced neurophysiologist, blinded to group identity for all participants, performed standardized nerve conduction studies at the clinical neurophysiology laboratory at Umeå University, Sweden. All measurements were conducted on the right leg, and included the amplitude and conduction velocity of the sural nerve. The conduction velocity of the peroneal nerve was also measured, but was not analysed further as it contains a larger proportion of motor neurons than the sural nerve. Thus, the sural nerve was deemed the most appropriate target with which to evaluate sensory nerve function.

Statistical analyses

Variables are summarized by numbers (*n*) and means (SD), according to the level of measurement. Differences between the NGT, IGT and Type 2 diabetes groups at baseline were analysed using ANOVA for continuous variables and the chi-squared test for categorical variables. Univariate mean changes over time were analysed using paired *t*-tests. For descriptive purposes, the bivariate association between HbA_{1c} and either the amplitude or conduction velocity of the sural nerve, were described using scatter plots together with Kendall's tau and Pearson's correlation coefficient. We have described the associations at both time points, as well as the relative change between 2004 and 2014.

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To take into account the repeated nature of the data and to account for possible confounders, regression analyses were then carried out. To handle non-normality, non-linearity and heteroscedasticity, we used the natural logarithms of HbA_{1c}, amplitude and conduction velocity at each time point. The associations between HbA_{1c} and amplitude or conduction velocity were then estimated using two different regression methods. First, we applied the fixed effects estimator that controls for confounders that are constant over time by using only the changes in HbA_{1c} that occurred within participants to estimate amplitude or conduction velocity. In addition, possible time-varying confounders were included in an adjusted analysis. Although the fixed effects estimator was robust with regard to confounders, a drawback was its low power because only within-individual variation between the two periods was utilized in the estimation. In a second analysis, therefore, we used the random effects estimator, which includes participant-specific effects as random effects. The random effects estimator relied on the assumption that individual-specific effects were uncorrelated to covariates, and both time-varying and time-invariant confounders had to be included in this analysis. If this assumption holds, the random effects estimator will have a higher power than the fixed effects estimator. In the event of this assumption not holding, the random effects estimator will be biased. A robust version of the Hausman test [18] was carried out to assess whether estimates from the fixed effects and random effects estimators differed [19].

To study the influence of missing data we used multiple imputations by chained equations [20]. All variables (dependent and independent) were included linearly in the imputation models (linear regression models). Imputation was carried out 20 times. The handling of missing data was restricted to survivors only, given that amplitude and conduction velocity cannot be defined for dead participants. The significance level was set to

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5% for all the analyses, with these performed in R 3.2.0 (R Core Team, 2015) using the software packages 'plm' [21] and 'mice' [20].

Results

Over the 10-year follow-up, the participants who had their amplitude measured on both occasions (*n*=74) were found to have experienced an average decrease in mean sensory nerve amplitude from 10.9 µV to 7.0 µV (*P*<0.001). This decline was most profound for participants who changed group from NGT to IGT, NGT to Type 2 diabetes, or IGT to Type 2 diabetes, with the caveat that relatively few participants changed group, which weakens any inference from this finding. Figure 2 shows mean changes according to group transition. The error bars represent standard errors. Individuals who transitioned to a group of lesser glucose tolerance (i.e. from NGT to IGT, NGT to Type 2 diabetes, or IGT to Type 2 diabetes; *n*=11) had, on average, an 8.3-µV decrease in sural amplitude, while those who remained in their group or improved (*n*=62) had on average a 3.2-µV decrease in sural amplitude. The difference in mean decrease in sural amplitude was significant (*P*=0.036). Regarding conduction velocity, a slight decrease was detected when analysing the mean change (Table 1), although this reduction was small and not statistically significant.

Association between HbA_{1c} and electrophysiology of the sural nerve

Analyses of the association between HbA_{1c} and the amplitude of the sural nerve response suggest that a 1% increase in HbA_{1c} is associated with an ~1% average decrease in amplitude (i.e. an increase from × mmol/mol 2004 to 1.01× mmol/mol 2014 is associated with 1% average decrease in amplitude). More specifically, the adjusted fixed effects estimate was –1.25 (95% CI –2.31, –0.18), while the adjusted random effects estimate was -0.84 (95% CI –1.32, –0.37; Table 2). Similar results were obtained when studying each time point

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separately (i.e. 2004 and 2014; Fig. 2). In addition, after accounting for HbA_{1c}, there was a 33% average decrease in sural nerve amplitude from 2004 to 2014 {100%*[exp (–0.41)–1]=33%; Table 2}.

Analyses of the association between HbA_{1c} level and sural nerve conduction velocity suggests a similar tendency to decline as levels of HbA_{1c} rise, but this (possible) association was weak (random effects estimate: –0.12, 95% CI –0.20, –0.05) and statistically not significant (fixed effects estimate: 0.002, 95% CI –0.22, 0.22; Table 2). Further, even though a crudely significant association between HbA_{1c} and conduction velocity was found in 2004 (Kendall's tau = –0.2, *P*=0.016), this association was weaker by 2014 (Kendall's tau = –0.15, *P*=0.059) and absent when studying the relative change (2004 to 2014; Fig. 2). When comparing the data of the participants who were lost to follow-up with the 2004 data of those who were followed up 2014, those lost to follow-up tended to have higher HbA_{1c} levels, diastolic blood pressure and serum creatinine levels; however, none of these differences were significant (*P*=0.12, *P*=0.14 and *P*=0.18, respectively). The only difference found was for cholesterol where those lost to follow-up had an average level of 4.9 mmol/l in 2004 compared with 5.4 mmol/l in 2004 for those who were followed up 2014 (*P*=0.017) .

When repeating the analysis on the imputed datasets, the associations between HbA_{1c} and amplitude or conduction velocity in Table 2 remained virtually the same or decreased slightly, e.g. the adjusted fixed effects changed from –1.25 to –1, but remained statistically significant (*P*=0.036).

Discussion

In the present 10-year follow-up of a cohort people with Type 2 diabetes, IGT or NGT (control group), we observed an association between HbA_{1c} and amplitude of the sural nerve,

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suggesting that a 1% increase in HbA_{1c} was associated with an ~1% average decrease in the amplitude of the sural nerve, irrespective of group classification. The finding was consistent over the 10-year study as well as at each time point (i.e. 2004 and 2014). The study population also showed a 33% average decrease in amplitude of the sural nerve over the follow-up period, after adjusting for HbA_{1c}. Similarly, conduction velocity significantly decreased, although its association with HbA_{1c} level was weak, indicating that degeneration and not demyelination is the prominent feature of neuropathy in Type 2 diabetes. Our findings corroborate the results of the study by Partanen *et al.* [12], in which the decreases in sensory (sural, peroneal, radial and median), and motor (peroneal and median) amplitudes were more pronounced than was the slowing of conduction velocity. To our knowledge, these are the only long-term follow-up studies to have been performed using participants with IGT and Type 2 diabetes. Whereas treatment recommendations may have changed, and additional pharmacological alternatives have become available since the study by Partanen *et al*., the course of Type 2 diabetes-associated neuropathy appears to follow the same pattern. Whether axonal degeneration reflects retraction, or is caused by apoptosis of the sensory neurons, as reported in animal experimentation studies [22–25], remains unclear.

Degeneration, whether axonal or via apoptosis, appears to be the main underlying cause of peripheral neuropathy in diabetes [26–31]. Previous studies have suggested that neuromuscular jitters and fibre density in single fibre electromyography assessment [32], and myelinated nerve fibre density [11], are both associated with HbA_{1c}. Together with our findings, this indicates that axonal degeneration and re-innervation processes are consistent with the metabolic control phase of the disease [11,32]. The decreased nerve amplitude, which is the most striking result of the present study, suggests that axonal degeneration may also be the primary mechanism behind hyperglycaemia-associated neuropathy. Conversely,

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reduced conduction velocity of the sural nerve, which relates to potential demyelination and remyelination processes, showed a relatively small decrease with time and no significant relationship to HbA_{1c} levels.

Type 1 and Type 2 diabetes exhibit marked differences in terms of their underlying neuropathology and pathogenetic mechanisms [28]; however, both share the feature of manifesting an initial metabolic phase that is amenable to metabolic correction (i.e. reversible) [11,28]. This metabolic phase is eventually displaced by a structural phase driven by degenerative changes through axonal atrophy and fibre loss that become increasingly refractile to metabolic intervention [28]. In the metabolic phase, the pathological process is dynamic with contemporaneous degenerative and regenerative processes [30]. In Type 2 diabetes, this initial metabolic phase comprises hyperglycaemia and an associated hyperlipidaemia and hypercholesterolaemia [28]. Hyperglycaemia leads to a shunting of excessive glucose through an activated polyol pathway which disrupts neural Na^+/K^+-ATPase, causing intra-axonal Na^+ accumulation [28]. In addition, impaired endoneurial blood flow, provoked by the decreased expression of endothelial nitric oxide synthase (eNOS) and nitric oxide (NO) activity, leads to endoneurial hypoxaemia [28] and chronic inflammation that may be associated with endothelial dysfunction and atherosclerosis [28]. Consequently, a deficit in nerve conduction seems to be the result of oxidative stress and decreased nerve perfusion [2,10,28,33]. Animal models suggest that these pathological mechanisms may be partially independent [28].

Axonal degeneration is milder in Type 2 than in Type 1 diabetes, but tends to be focal in older people with Type 2 diabetes, suggesting a more prominent vascular genesis [26]. A further characteristic feature of diabetic neuropathy is blunted fibre regeneration, probably as a consequence of an impaired immune response and synthesis of local neurotrophic factors

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[26].

Other studies have shown an association between plasma glucose and corneal nerve fibre density and branching [34], neuromuscular jitters [32], and myelinated fibre density [11,32], while plasticity and the regenerative ability of peripheral nerve fibres remains intact [7,28,30]. In other words, glucose control has an effect on the long-term state of peripheral nerves in both people with IGT and those with Type 2 diabetes [11,32,34].

There is increasing evidence that peripheral neuropathy may start as early as during prediabetes [3–6]. This and the present findings suggest the importance of prompt action, not only for those diagnosed with Type 2 diabetes, but also for those in the prediabetes (IGT) stage [11,32,34]. For improved long-term outcomes, it would seem that early intervention is important, when some degree of biological plasticity still remains and nerve regeneration is possible [11,32, 34].

An interesting observation is that the association between HbA_{1c} and sural nerve amplitude was also found among participants with NGT and those with IGT. Indeed, participants initially categorized in the NGT or IGT group, who were subsequently diagnosed with Type 2 diabetes during follow-up, showed a more profound decrease in amplitude than those participants diagnosed with Type 2 diabetes at baseline who were treated and lived with diabetes throughout the 10-year follow-up. This finding highlights the importance of early detection, and the timely treatment of people in the prediabetes stage, as this appears to be a phase when much can be gained by intervention.

The present study has some notable strengths, including a uniquely long follow-up period, together with assessments that were carried out by the same staff, in the same fashion, at both time points. Study limitations include the high drop-out rate, which requires us to interpret our follow-up data with caution. Participants who had died or who declined participation in

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2014 had a tendency to have higher than average HbA_{1c} levels, diastolic blood pressure and serum creatinine levels in 2004; however, none of these differences was statistically significant. The only difference found was for cholesterol, where the participants who had died or declined participation had lower than average levels compared with the 2004 value of those who were followed up.

In conclusion, HbA_{1c} level was found to be associated with the amplitude of the sural nerve, at all levels of HbA_{1c}. Early detection is therefore likely to be important for the prevention of neuropathy in people with IGT as well as those with Type 2 diabetes.

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Competing interests

None declared.

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Prior presentation

Aspects of this study were presented at the 76th Scientific Sessions of the American Diabetes Association, New Orleans, LA, 10–14 June 2016, and the 16th World Congress on Pain, Yokohama, Japan 26–30 September 2016.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Comparison of the participants followed up in 2014 (2004 values) with participants lost-to-follow-up.

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Table 1 Baseline characteristics of the study population in 2004, consisting of people with normal and impaired glucose tolerance and Type 2 diabetes

	NGT	IGT	Type 2 diabetes	*P*	All
n	39	29	51		119
Women, *n* (%)	20 (51.3)	14 (48.3)	21 (41.2)	0.615	55 (46.2)
Age, years	61.1 (0.6)	61.4 (0.8)	61.0 (1.3)	0.123	61.1 (1.0)
Diabetes duration, years	na	na	7.0 (5.7)	na	7.0 (5.7)
BMI, kg/m^2	26.0 (3.6)	26.9 (5.4)	29.4 (4.6)	0.002	27.7 (4.8)
HbA_{1c}, mmol/mol	34.9 (2.9)	36.4 (4.0)	58.1 (14.6)	<0.001	45.2 (14.9)
HbA_{1c}, %	5.3 (0.3)	5.5 (0.4)	7.5 (1.3)	<0.001	6.3 (1.4)
Diastolic blood pressure, mmHg	76.2 (7.4)	75.4 (11.2)	75.8 (7.1)	0.921	75.8 (8.3)
S-creatinine, µmol/l	75.5 (14.8)	73.5 (14.1)	76.0 (28.6)	0.874	75.2 (21.5)
GFR, ml/min	93.2 (20.6)	99.1 (25.0)	110.5 (27.7)	0.005	102.0 (25.8)
Total cholesterol, mmol/l	5.8 (0.8)	5.4 (0.9)	4.7 (0.7)	<0.001	5.2 (0.9)
Triglycerides, mmol/l	1.3 (0.6)	1.3 (0.5)	1.7 (0.9)	0.014	1.5 (0.8)
Amplitude sural nerve, uV	11.1 (6.6)	11.0 (6.1)	9.6 (6.6)	0.507	10.4 (6.5)

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Conduction velocity sural nerve, m/s	48.5 (6.5)	47.4 (5.9)	45.5 (5.5)	0.013	46.9 (6.1)	

IGT, impaired glucose tolerance; NGT, normal glucose tolerance.

Values are means (SD), unless otherwise indicated.

Table 2. Linear fixed effects and linear random effects estimation of the associations between logarithmic HbA_{1c}, logarithmic amplitude (µV), and logarithmic conduction velocity (m/s) of the sural nerve, in participants with normal and impaired glucose tolerance, and in people with Type 2 diabetes

Dependent variable	**Log amplitude, µV**				**Log conduction velocity, m/s**			
Estimator	**Crude fixed effects estimate (95% CI)**	**Adjusted fixed effects estimate (95% CI)**	**Crude random effects estimate (95% CI)**	**Adjusted random effects estimate (95% CI)**	**Crude fixed effects estimate (95% CI)**	**Adjusted fixed effects estimate (95% CI)**	**Crude random effects estimate (95% CI)**	**Adjusted random effects estimate (95% CI)**
Model:	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**

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log HbA_{1c} (mmol/mol)	–0.80 (-1.65, .05)	–1.25* (–2.31, –0.18)	–0.79*** (–1.19, –0.39)	–0.84*** (–1.32, –0.37)	0.02 (–0.18, 0.21)	0.002 (–0.22, 0.22)	–0.10** (–0.17, –0.04)	–0.12** (–0.20, –0.05)
Year (2014)	–0.41*** (–0.54, –0.27)	–0.14 (–0.38, 0.10)	–0.39*** (–0.51, –0.27)	–0.41*** (–0.59, –0.24)	–0.04* (–0.07, –0.01)	–0.04 (–0.09, 0.01)	–0.03* (–0.06, –0.002)	–0.04** (–0.08, –0.01)
eGFR (ml/min)		0.02** (0.01, 0.03)		–0.001 (–0.01, 0.01)		–0.001 (–0.003, 0.001)		–0.001** (–0.002, –0.0003)
Cholesterol (mmol/l)		–0.002 (–0.14, 0.14)		–0.04 (–0.14, 0.05)		–0.03 (–0.06, 0.0000)		–0.001 (–0.02, 0.02)
Triglycerides (mmol/l)		0.04 (–0.18, 0.25)		0.06 (–0.08, 0.20)		0.03 (–0.02, 0.07)		0.003 (–0.02, 0.03)
BMI (kg/m^2)		–0.03 (–0.10, 0.05)		0.002 (–0.03, 0.03)		0.01 (–0.01, 0.02)		0.01** (0.003, 0.01)

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Diastolic blood pressure (mmHg)		–0.0004 (–0.02, 0.02)		–0.003 (–0.01, 0.01)		–0.003 (–0.01, 0.0004)		–0.001 (–0.003, 0.001)
Sex (male)				–0.45** (–0.72, –0.18)				–0.05* (–0.10, –0.01)
Age (years)				0.01 (–0.12, 0.14)				–0.003 (–0.02, 0.02)
Constant			5.09*** (3.57, 6.61)	5.32 (–3.23, 13.87)			4.23*** (3.98, 4.48)	4.52*** (3.21, 5.84)
Observations	188	173	188	173	187	172	187	172
Hausman test	Model 1 vs Model 3		Model 2 vs Model 4***		Model 5 vs Model 7		Model 6 vs Model 8	

eGFR, estimated creatinine clearance rate using the Cockcroft–Gault formula. *P<0.05; **P<0.01; ***P<0.001.

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FIGURE 1 Flowchart describing the study. IGT, impaired glucose tolerance; NGT, normal glucose tolerance.

FIGURE 2 Mean change in sural nerve amplitude, conduction velocity, and HbA_{1c} level in participants with normal glucose tolerance (NGT) and impaired glucose tolerance (IGT), and in participants with Type 2 diabetes, between 2004 and 2014, according to group transition.

FIGURE 3 Crude associations between HbA_{1c} (mmol/mol), amplitude (µV), and conduction velocity (m/s), in 2004, and 2014, and their relative changes. Fitted curves are from crude simple linear regression analyses, regressing the logarithm of the outcome (*y*-axis) on the logarithm of the independent variable (*x*-axis; e.g. curves C and F are derived from Models 1 and 5 in Fig. 2) and then transforming back to the original scale.

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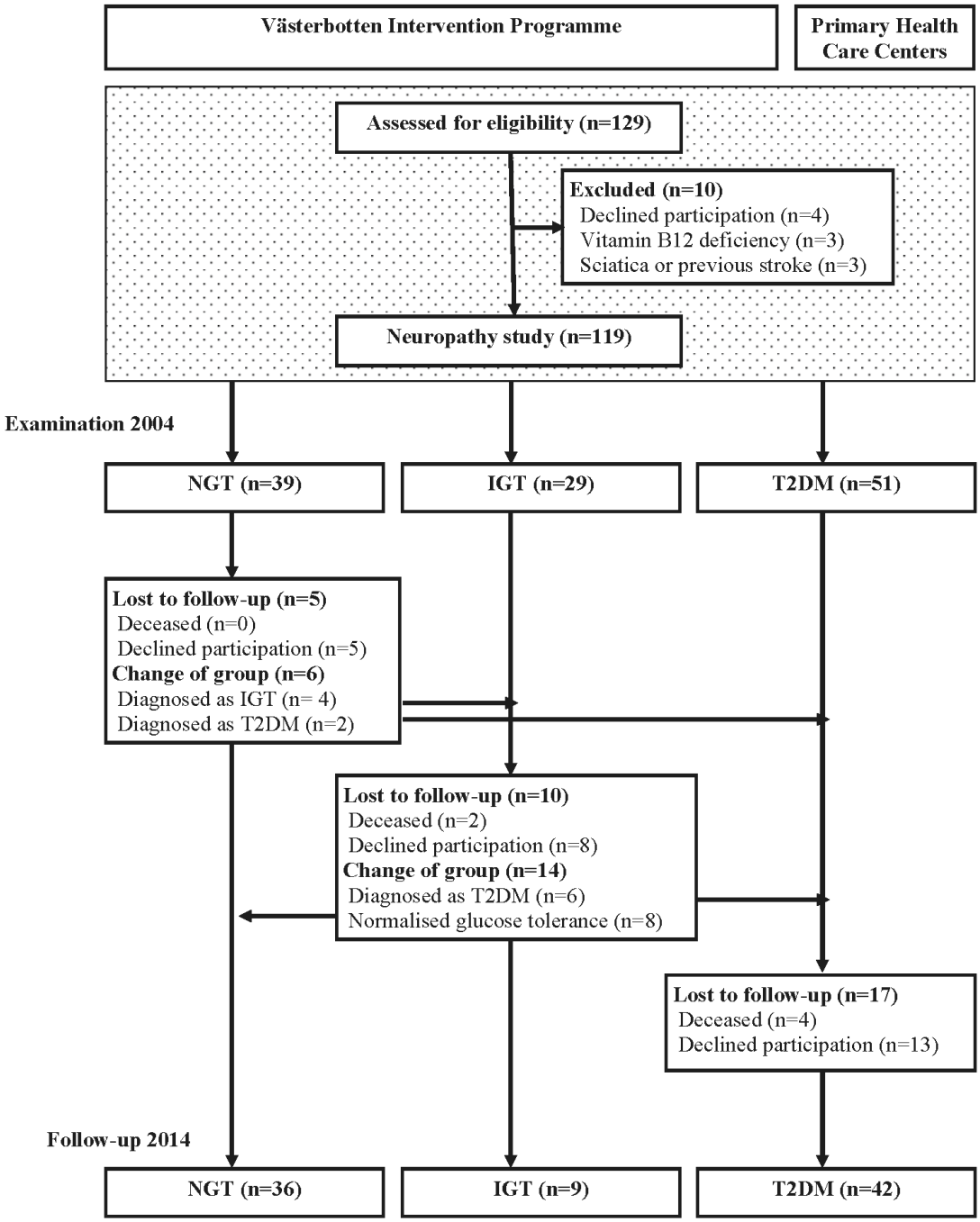

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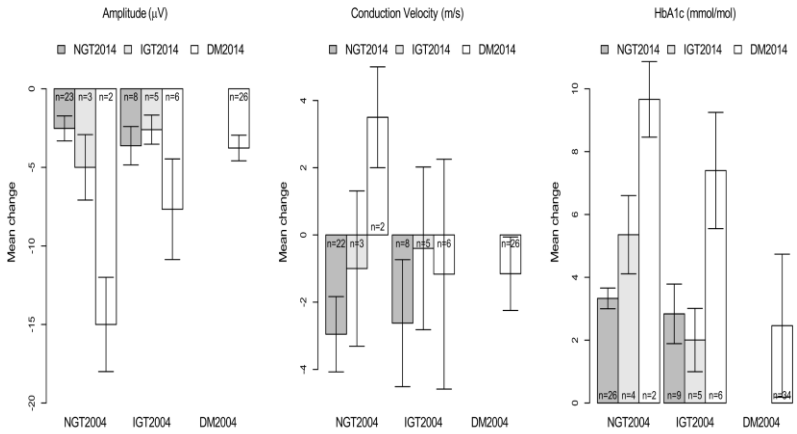

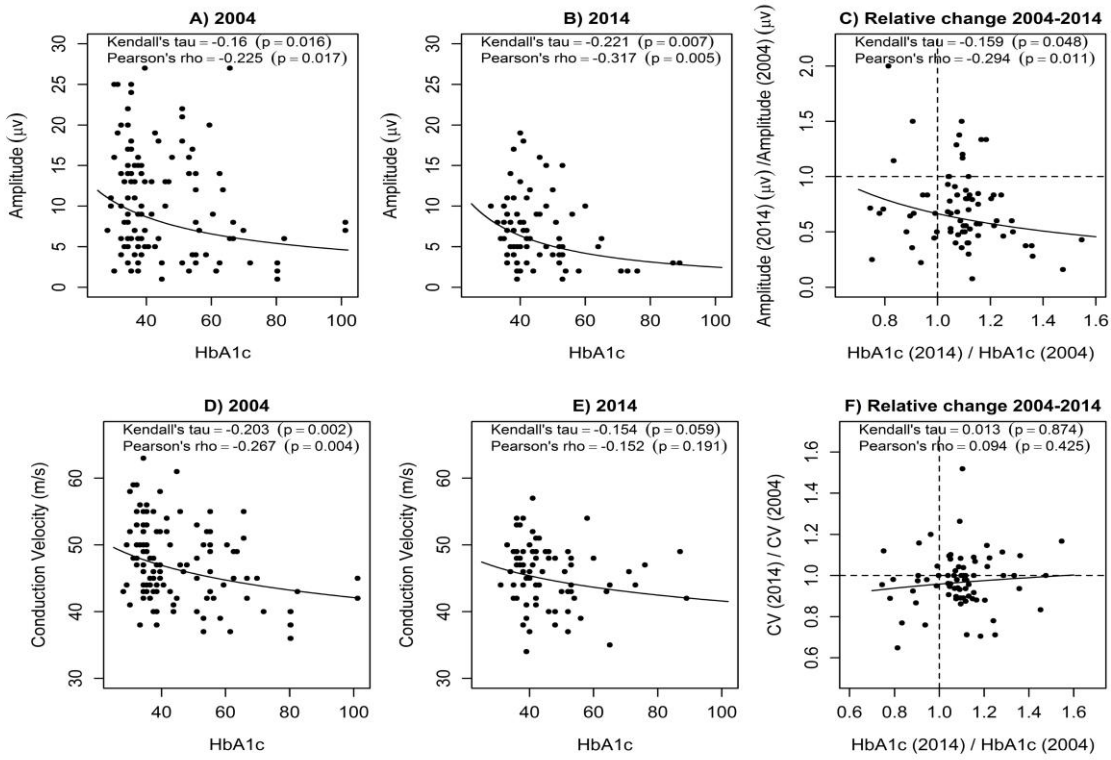

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